

FIXING THE GAPS

A Stakeholder-Driven Power BI Dashboard for Resource Allocation and School Performance in Pakistan

Business Intelligence & Advanced Data Analytics — Capstone Project

Prepared for KP-IT Board Course Evaluation

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1. Business Problem

Why this BI solution is needed

1. Education data is fragmented across performance, infrastructure, geography and contextual indicators.
2. Decision makers need to identify where resource gaps are most severe, not only view national averages.
3. Traditional tables report numbers but do not clearly prioritize districts for intervention.
4. The solution converts available data into an interactive decision support dashboard.

2013-2016

Primary analysis period

Province → City

Geographic analysis

Evidence → Action

Decision-support
focus

2. Project Objectives

1. Measure education, learning and retention performance across regions.
2. Identify infrastructure deficits involving electricity, drinking water, toilets, classrooms and teachers.
3. Highlight gender enrollment disparities and high-risk regions.
4. Connect findings to specific, actionable government interventions.
5. Provide executives with a consistent, interactive BI reporting experience.

3. Dataset & Data Sources

Core educational and contextual indicators

Education

Enrollment, completion & performance

Infrastructure

Facilities, classrooms & teachers

Geography

Province, city/district & population

Context

Budget, economic & security indicators

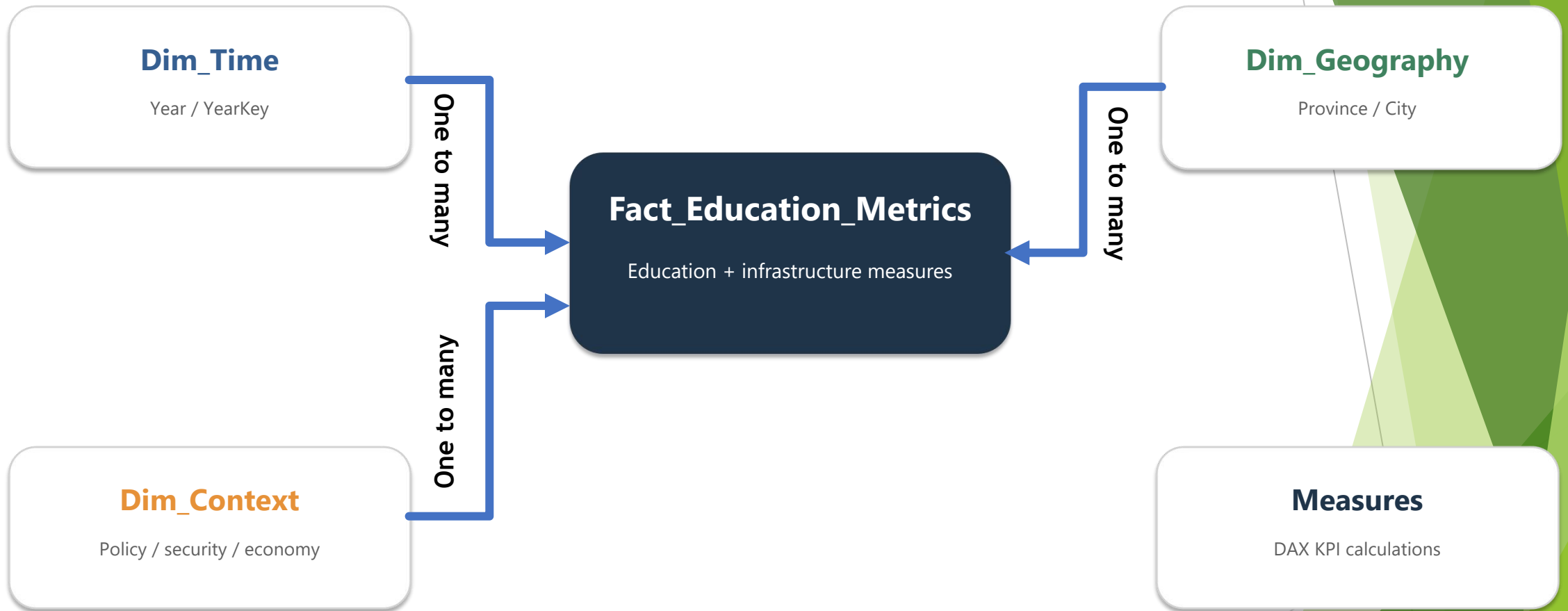
- **Main dataset:** Consolidated Educational / Pakistan Education Performance Dataset.
- **Analysis period:** 2013–2016.
- Key fields include boys' and girls' enrollment, facility availability, single-teacher and single-classroom indicators.
- Context fields include Global Terrorism Index drone attacks, bomb blasts, economic growth and educational budget spend of GDP.

4. Data Preparation & Quality

1. Reviewed column names, data types, blanks/nulls and duplicate records.
2. Converted percentage and numeric fields to appropriate data types.
3. Standardized geographic and categorical values.
4. Created separate Time, Geography and Context dimensions.
5. Validated key measures using direct aggregation, filtered visual checks and cross-visual comparison.
6. Reviewed Year and Province filter behavior and corrected relevant model relationships.
7. Applied consistent formatting and conditional formatting for risk indicators.

5. Data Model

Star-schema approach



Dimension tables filter the central fact table; measures respond to report filter context.

6. KPI Framework

Gender Enrollment Gap

Boys enrollment – girls enrollment

Utility Deficit Rate

No-facility schools / total schools

Teacher Strain

Single-teacher schools / primary schools

Classroom Strain

Single-classroom schools / primary schools

Education Score

Overall education performance

Learning Score

Learning performance indicator

Retention Score

Retention performance indicator

Overall Risk Score

Combined risk indicator

7. Dashboard Architecture

Executive Governance Panel

High-level KPIs, education performance and governance view

Executive Risk & Intervention

Critical regions, risk scoring and priority interventions

District Infrastructure Audit

Province/city infrastructure gaps and drill-through analysis

Policy Impact & Socio-Economic

Education outcomes in relation to policy and socio-economic context

Dashboard 1 — Executive Governance Panel

Pakistan Education Performance | Government Education Governance



FIXING THE GAPS - (Executive Governance Panel)

Pakistan Education Performance | Government Education Governance

Year

All

Province, City

All



Clear Filters

Total Schools



608K

Primary Schools



489K

Secondary Schools



120K

Education Score



61.15

Critical Gender Gap Regions



8

Executive Governance

Risk & Intervention

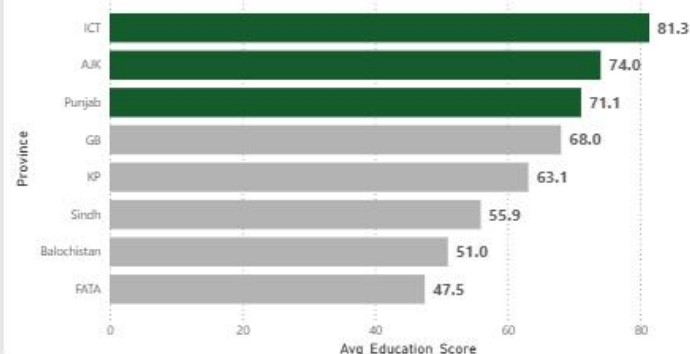
Infrastructure Audit

Infrastructure Analysis

Policy Impact & Socio...

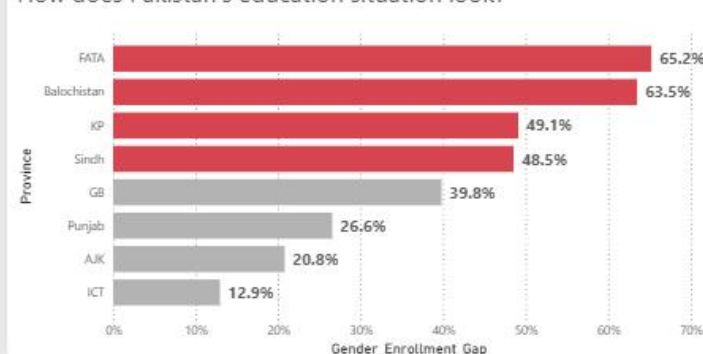
Socio Economic...

Average Education Score by Province

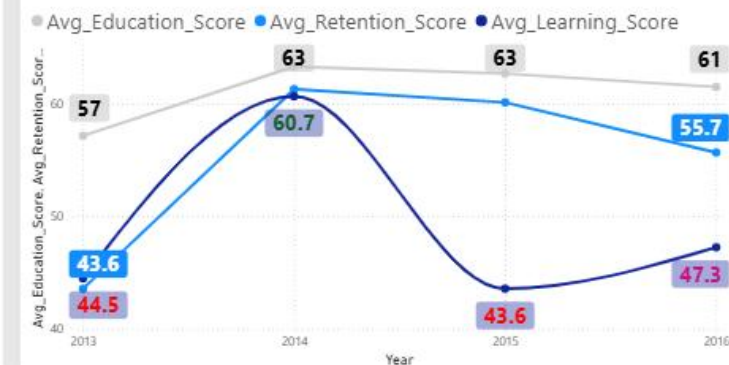


Gender Enrollment Gap by Province

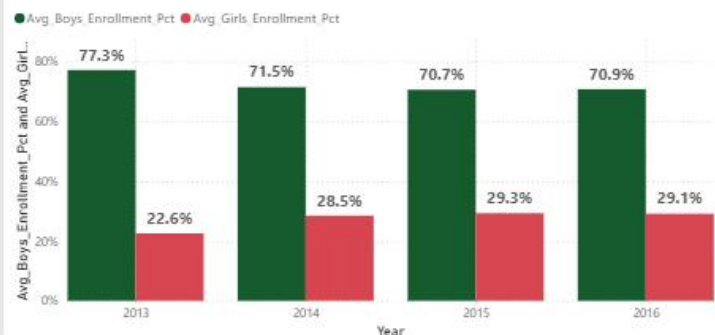
How does Pakistan's education situation look?



Education, Learning & Retention Trends



Boys vs Girls Enrollment Trend



Regional Infrastructure Policy Context

Comparing essential facility coverage and infrastructure gaps across provinces and districts

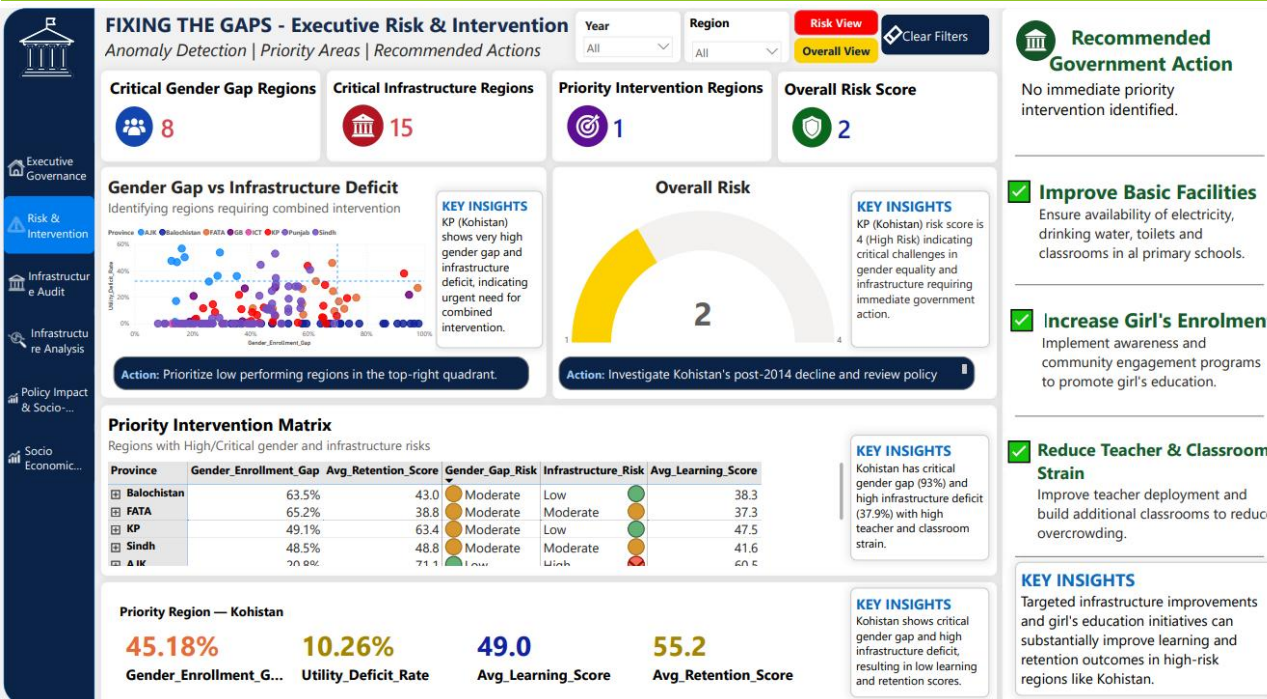
Province	Total_Schools	Total_Primary_Schools	Total_Secondary_Schools	Avg_Education_Score	Gender_Enrollment_Gap	Gender_Gap_Risk
ICT	1564	764	800	81.34	12.95%	Low
AJK	23940	16808	7132	74.00	20.79%	Low
Punjab	211944	151412	60532	71.05	26.55%	Low
GB	5100	3188	1912	68.04	39.79%	Low
KP	112712	92088	20624	63.11	49.11%	Moderate
Sindh	177584	160544	17040	55.93	48.53%	Moderate
Balochistan	51468	43240	8228	50.98	63.50%	Moderate
FATA	24044	20456	3588	47.50	65.23%	Moderate

Source: Pakistan Education Performance Dataset | Kaggle | Historical district-level data, 2013–2016

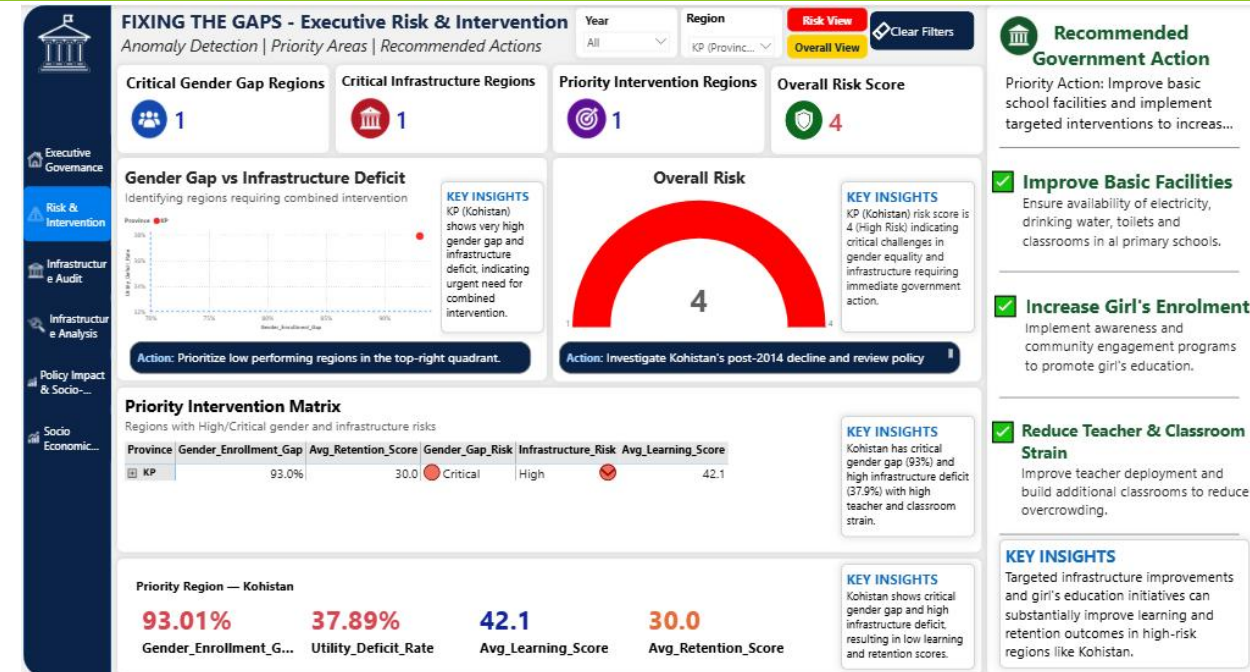
Dashboard 1 — Executive Risk & Intervention

Risk prioritization, critical regions and recommended government action

Overall View



Risk View



10. Priority Intervention: Kohistan

93.0%

Gender Enrollment Gap

37.89%

Utility Deficit

28.6%

Single-Teacher Strain

20.5%

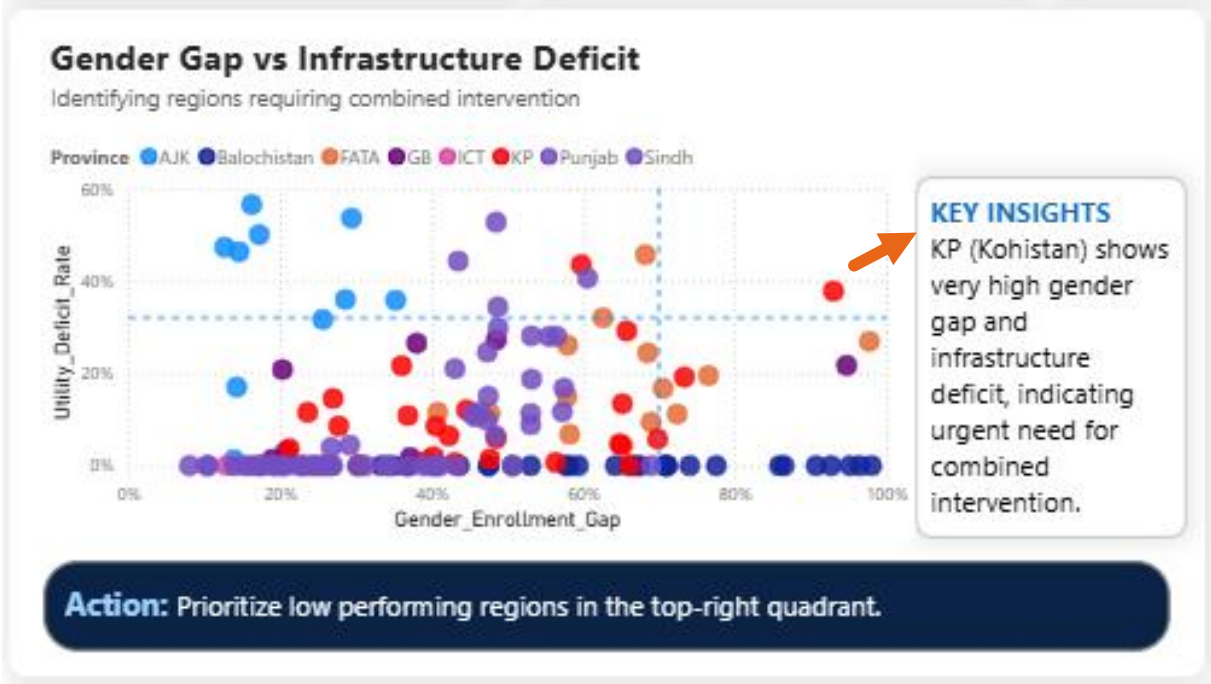
Single-Classroom Strain

42.1

Learning Score

30.0

Retention Score



Why it matters

Kohistan is supported by multiple risk indicators simultaneously, making it a strong priority case rather than one based on a single KPI.



Recommended Government Action

No immediate priority intervention identified.



Improve Basic Facilities

Ensure availability of electricity, drinking water, toilets and classrooms in al primary schools.



Increase Girl's Enrolment

Implement awareness and community engagement programs to promote girl's education.



Reduce Teacher & Classroom Strain

Improve teacher deployment and build additional classrooms to reduce overcrowding.

Dashboard 2 — District Infrastructure Audit

Infrastructure, Facilities & Staffing Risk



Executive
Governance

Risk &
Intervention

Infrastructure
Audit

Infrastructure
Analysis

Policy Impact
& Socio...

Socio
Economic...

DISTRICT INFRASTRUCTURE AUDIT

Infrastructure, Facilities & Staffing Risk

Year

All

Province

All

Clear Filters

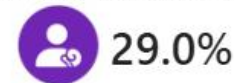
Utility Deficit Rate



Average Drinking Water



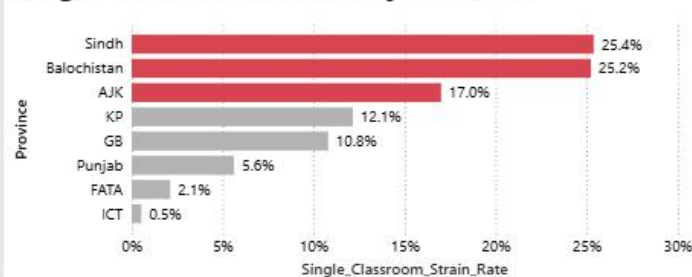
Single-Teacher Strain



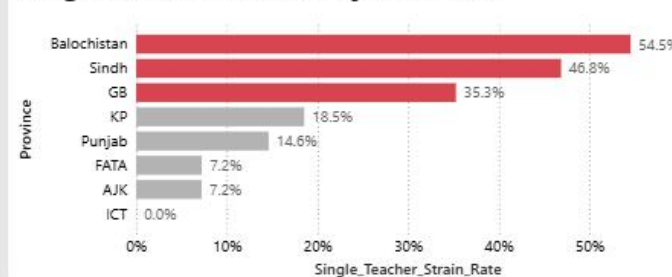
Single-Classroom Strain



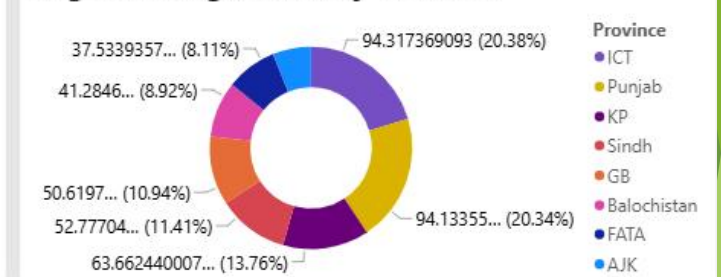
Single-Classroom Strain by Province



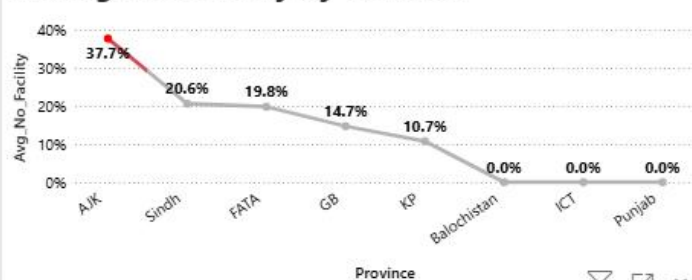
Single-Teacher Strain by Province



Avg_Drinking_Water by Province



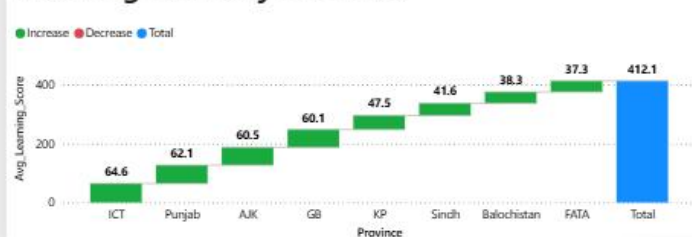
Average No Facility by Province



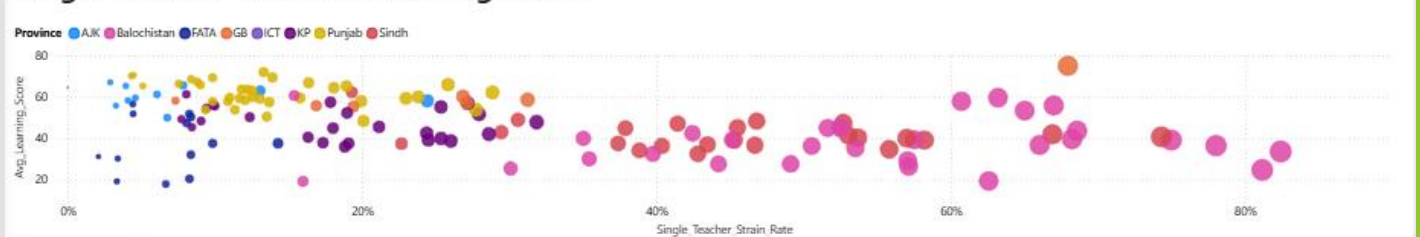
District Infrastructure Audit

Province	Single_Classroom_Strain_Rate	Avg_Electricity	Avg_Drinking_Water	Avg_Toilet	Avg_Building_Condition	Infrastructure_Risk
AKJ	17.0%	14.92	28.38	30.08	34.63	High
Balochistan	25.2%	26.55	41.28	22.34	21.05	Low
ICT	0.5%	98.33	94.32	96.83	46.21	Low
KP	12.1%	53.16	63.66	76.24	70.75	Low
Punjab	5.6%	75.35	94.13	87.66	77.17	Low
FATA	2.1%	40.48	37.53	37.27	26.44	Moderate
GB	10.8%	43.93	50.62	47.87	24.10	Moderate

Learning Score by Province



Single-Teacher Strain vs Learning Score



Dashboard 2 — District Infrastructure Audit

Infrastructure, Facilities & Staffing Risk

ICT

98.33 Electricity • 94.32 Drinking Water • 96.83 Toilets

Punjab

75.35 Electricity • 94.13 Drinking Water • 87.66 Toilets

Sindh

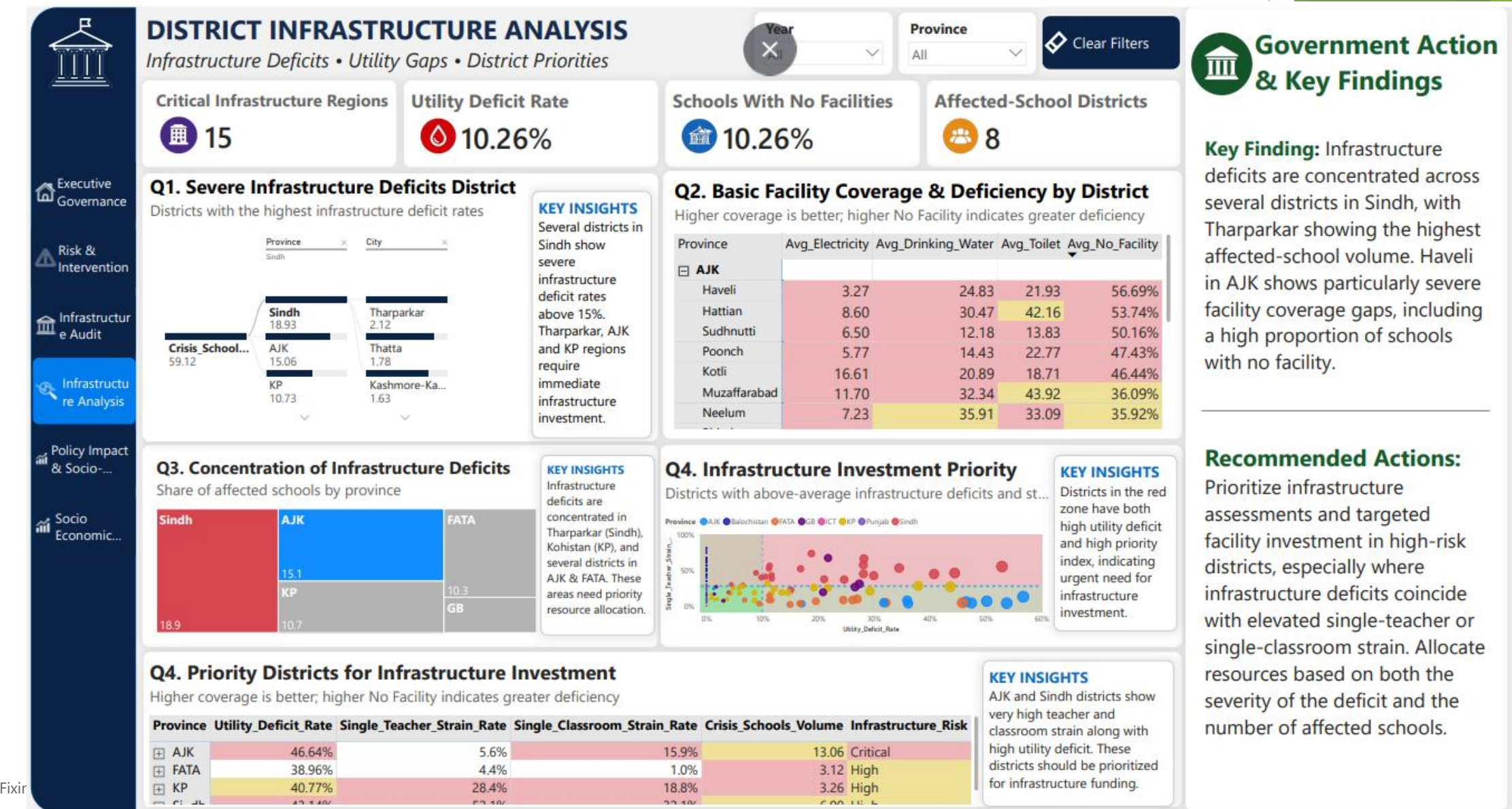
44.19 Electricity • 52.78 Drinking Water • 59.04 Toilets

Balochistan

26.55 Electricity • 41.28 Drinking Water • 22.34 Toilets

Dashboard 2 — District Infrastructure Analysis

Infrastructure deficits, utility gaps and district investment priorities



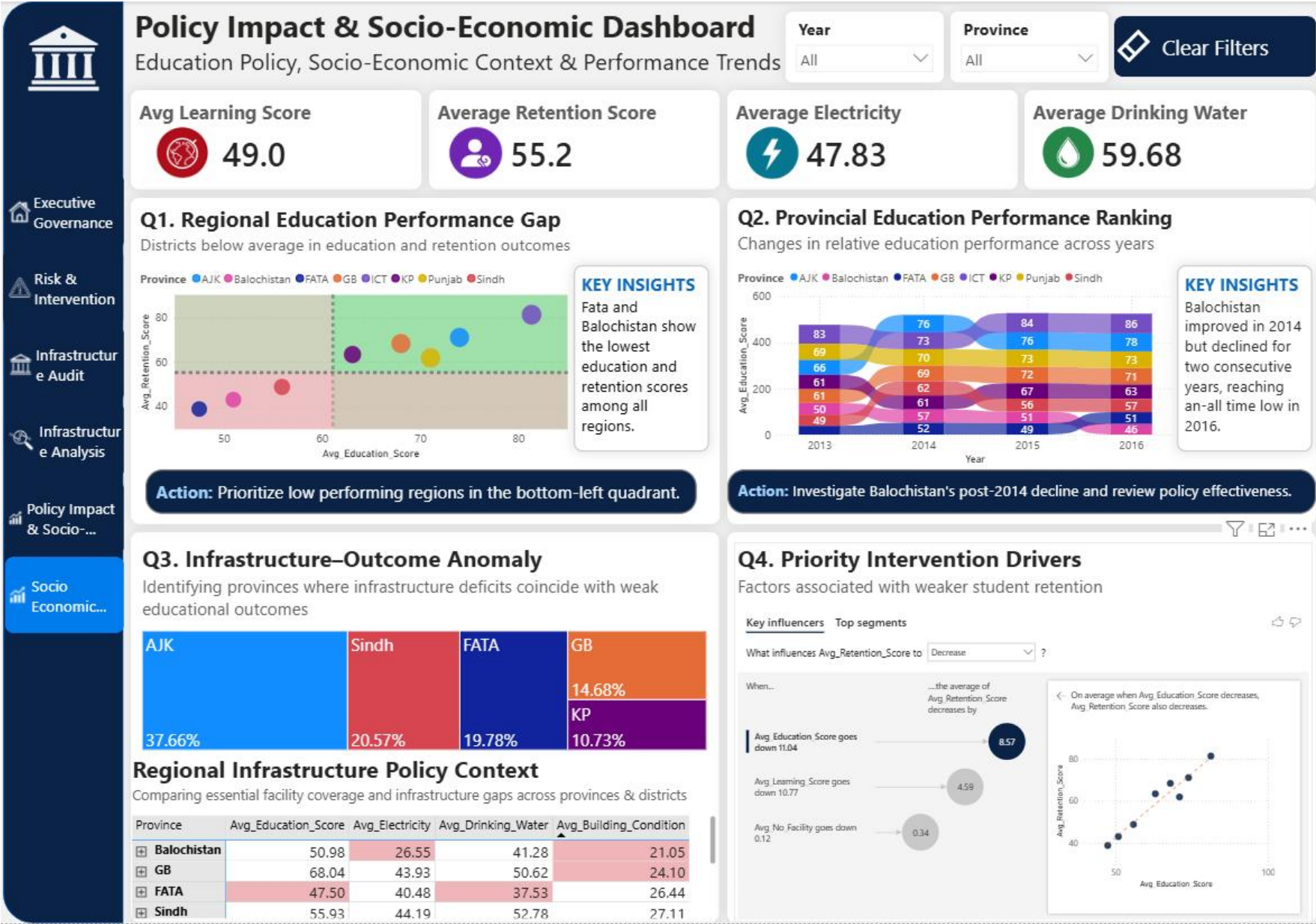
Dashboard 3 — Policy Impact & Socio-Economic

Education outcomes viewed alongside policy and socio-economic context



Dashboard 4 — Policy Impact & Socio-Economic Analysis

Interactive analysis view with anomaly and intervention-oriented visuals



9. Trends & Anomalies

AJK: 66.2 → 78.1

Education score, 2013 to 2016

**Sindh: 48.9 → 61.7
→ 56.2 → 56.9**

2013–2016; 2014 gain not sustained

**Balochistan: 49.8 → 57.2
→ 50.8 → 46.1**

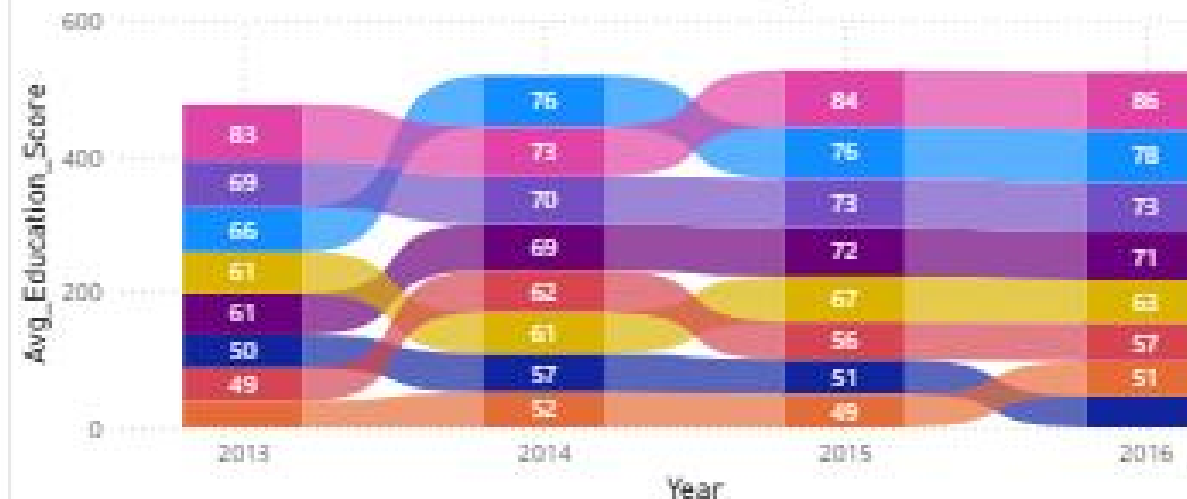
Improvement reversed after 2014

- AJK shows sustained improvement across the analysis period.
- Sindh shows a sharp 2014 improvement followed by a decline and slight recovery.
- Balochistan's post-2014 decline is a key anomaly requiring investigation.
- Sudhnoti shows that high infrastructure deficit does not automatically imply low retention, so local context matters.

Q2 — Provincial Education Performance Ranking

Changes in relative education performance across years

Province ● AJK ● Balochistan ● FATA ● GB ● ICT ● KP ● Punjab ● Sindh



KEY INSIGHTS

Balochistan improved in 2024 but declined for two consecutive years, reaching an all time low in 2016.

Action: Investigate Balochistan's post-2014 decline and review policy effectiveness.

11. District Infrastructure Evidence

Tharparkar

52.89% No Facility

33.1 Retention

Thatta

44.42% No Facility

24.3 Retention

Haveli

56.69% No Facility

55.6 Retention

Hattian

53.74% No Facility

55.6 Retention

Sudhnoti

50.16% No Facility

85.4 Retention

Tor Ghar

43.64% No Facility

51.0 Retention

KEY INSIGHTS

Similar infrastructure deficits can coexist with very different outcomes, so interventions should be district-specific.

District Infrastructure Audit

Province	Single_Teacher_Strain_Rate	Single_Classroom_Strain_Rate	Avg_Electricity	Avg_Drinking_Water	Avg_Toilet	Avg_Building_Condition
⊕ Sindh	46.8%	25.4%	44.19	52.78	59.04	27.11
⊕ Balochistan	54.5%	25.2%	26.55	41.28	22.34	21.05
⊕ AJK	7.2%	17.0%	14.92	28.38	30.08	34.63
⊕ KP	18.5%	12.1%	53.16	63.66	76.24	70.75
⊕ GB	35.3%	10.8%	43.93	50.62	47.87	24.10
⊕ Punjab	14.6%	5.6%	75.35	94.13	87.66	77.17
⊕ FATA	7.2%	2.1%	40.48	37.53	37.27	26.44
⊕ ICT	0.0%	0.0%	00.00	00.00	00.00	00.00

12. Policy Impact & Socio-Economic Analysis

1. Compares education outcomes with contextual indicators such as educational budget, economic growth and security-related measures.
2. Educational budget is treated as a national-level indicator; therefore it responds to Year but is not expected to change with a Province slicer.
3. The analysis reveals associations and patterns; it does not claim that one contextual variable directly causes an education outcome.
4. Province and district comparisons show where national policy context is associated with different local outcomes.

National Budget

Year-sensitive context

Regional Outcomes

Province/city comparison

Decision Lens

Policy → evidence → action

13. Recommended Government Actions

HIGH

Prioritize integrated interventions in Kohistan and other critical regions: facilities + teachers + classrooms + gender participation.

HIGH

Target Sindh and Balochistan districts where high no-facility rates coincide with weak learning and retention.

MEDIUM

Study high-performing/improving regions such as ICT and AJK to identify practices that can be adapted elsewhere.

MEDIUM

Investigate reversals such as Balochistan after 2014 rather than evaluating policy only from a single year's result.

15. Performance Optimization

225.40 → 183.92 ms

District page average processing time

3,381 → 2,391 ms

District page total processing time

214.78 → 202.28 ms

Executive Governance average

- Reduced unnecessary visual processing and consolidated report elements.
- Used Performance Analyzer to identify expensive visuals/queries.
- Kept analytical visuals required for decision-making while improving responsiveness.
- Before/after performance evidence was documented in the project workbook.

16. Expected Business & Policy Impact

Faster

Identify priority regions

Clearer

Translate KPIs into insights

Targeted

Allocate resources by need

Repeatable

Monitor trends over time

- Reduce the time required to locate high-risk districts across multiple indicators.
- Support evidence-based resource allocation instead of relying only on aggregate provincial averages.
- Create a repeatable reporting structure for monitoring education performance.
- Improve communication between analytical evidence and policy recommendations.

17. Conclusion

Data → Gaps → Evidence → Priority Regions → Government Action

1. The solution combines education, infrastructure, geography and contextual indicators into one decision-support environment.
2. The analysis demonstrates significant regional variation and identifies specific high-priority districts.
3. Kohistan is a clear multi-indicator intervention case; Sindh and Balochistan contain districts requiring targeted support.
4. The dashboard helps stakeholders move from observing performance to deciding where action is needed.

THANK YOU